Lesson 8.5 - The Standard Error of the Mean

> **Video:** Standard Error of the Mean by The Organic Chemistry Tutor **Chapter:** 8 - Normal Probability Distributions

Formula

The **standard error of the mean (SE)** is the standard deviation of the sampling distribution of $\bar{x}$. It measures how much sample means vary from sample to sample.

You know	Standard error
The **sample** SD s	$\text{SE} = \dfrac{s}{\sqrt{n}}$
The **population** SD σ	$\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$

Example 1 - from sample data

The ages of 5 randomly selected college students are 18, 21, 19, 20, 26.

(a) Sample standard deviation

Mean: $\bar{x} = \dfrac{\sum x}{n} = \dfrac{104}{5} = 20.8$

x	$x - 20.8$	squared
18	−2.8	7.84
21	0.2	0.04
19	−1.8	3.24
20	−0.8	0.64
26	5.2	27.04
Sum		**38.80**

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} = \sqrt{\frac{38.8}{5 - 1}} = \sqrt{9.7} \approx 3.114$$

(b) Standard error

$$\text{SE} = \frac{s}{\sqrt{n}} = \frac{3.114}{\sqrt{5}} \approx 1.393$$

Example 2 - from the population SD

At a university, student ages have mean 20.5 and **population** SD $\sigma = 0.8$.

(a) Samples of $n = 25$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{25}} = \frac{0.8}{5} = 0.16$$

(b) Samples of $n = 100$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{100}} = \frac{0.8}{10} = 0.08$$

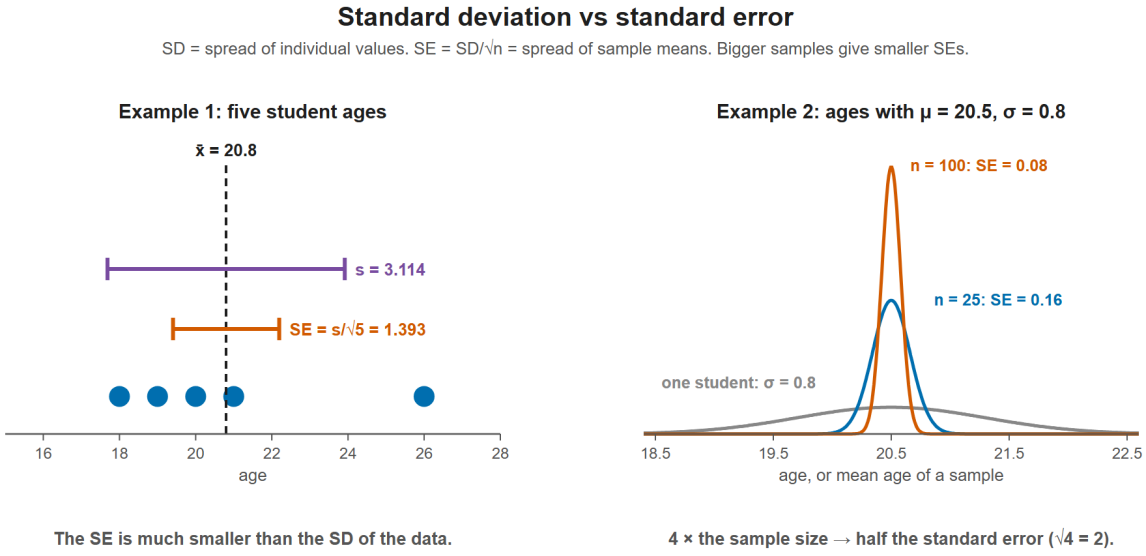
Quadrupling n halves the standard error, because $\sqrt{4} = 2$.

What it looks like

Plot the means of many samples to get the **sampling distribution of the mean**, a bell curve centered at μ.

- With $n = 25$ it is wider and shorter.
- With $n = 50$ (or 100) it is **narrower and taller**.

A smaller SE means the sample means cluster more tightly around μ, so any one sample mean is a **more accurate** estimate.



Left: the SD describes how spread out the five ages are, while the SE describes how much the mean of five ages would vary. Right: going from n = 25 to n = 100 halves the SE, so the curve of sample means gets twice as narrow.

Key takeaways

- SE = SD ÷ $\sqrt{n}$. Use s from a sample or σ from the population.
- The SD describes spread among **individual values**. The SE describes spread among **sample means**.
- SE ∝ $1/\sqrt{n}$: a larger sample gives a smaller SE and a more precise estimate.